Birla Institute of Technology and Science, Pilani
Hyderabad Campus
Semester I, Academic Year 2026–2027
CS F425 – Deep Learning

(Evaluative & Take-Home) Project Assignment – II

Zero-Shot PCB Segmentation: Compact Student Models vs. Teacher Pseudo-Labels

Total: 60 Marks | Course Weightage: 20%

Instructor In-Charge	Prof. Sk Aziz Ali skaziz.ali@hyderabad.bits-pilani.ac.in
Assignment Release	**3rd November 2026**
Submission Deadline	**6th December 2026** (No extension!)

MARK DISTRIBUTION: Question 1 – Semantic Segmentation: **24** + Question 2 – Instance Segmentation: **30** + Submission & Reproducibility: **6** = **60 marks**.

1. General Instructions (Continued from Assignment–I)

- Work only in the **pre-assigned group**; **ONE** final solution and **ONE** submission email per group. Group membership remains unchanged.
- Create/use a **PRIVATE (NOT public)** GitHub repository named `CSF425_2026_XX` (where XX is the group ID). Maintain `src/`, `scripts/`, `results/`, `checkpoints/`, `README.md`, and `config.yml` (or `environment.yml`). Clearly distinguish Assignment–II files from Assignment–I.
- Only one member must email the Instructor with subject `[CSF425_2026_XX_Assignment-II]` and exactly one file `CSF425_2026_XX_Assignment-II.zip`. Use the correct group ID; do not submit multiple versions.
- GitHub, code, logs, images, and documents must be sufficient for a non-expert to reproduce results. Do not include dataset archives, unnecessarily large temporary data, or SSH credentials in the repository/ZIP.
- The original rule for external source data remains: at most **1.5× the original labeled training-set size** without data-volume grading decay. Standard on-the-fly augmentation does not count as adding distinct external images. *The frozen student model must be trained and selected on source-domain data only. Teacher-generated DSLR masks are evaluation pseudo-labels, not target training labels for the graded zero-shot track.*
- Maintain proper citations for external data, pretrained weights, adapted source code, and foundational papers. Do not manually edit evaluation predictions, hide teacher-label failures, or cherry-pick test images. **The final model you submit must NOT cross the parameter count of YOLO26-M-seg.**

1.1 HPC / Kaggle access and security

Each group receives its assigned account and SSH keys. Your private key `csisdl_XX` **must never be shared or committed**. Run the following on your own machine:

```
chmod 600 csisdl_XX
```

```
ssh -X -i ./csisdl_XX csisdl_XX@hpc.bits-hyderabad.ac.in
```

If the key is already configured in your SSH agent/configuration, the shorter command is:

```
ssh -X csisdl_XX@hpc.bits-hyderabad.ac.in
```

Optional alias in `~/.bashrc` (replace the absolute key path):

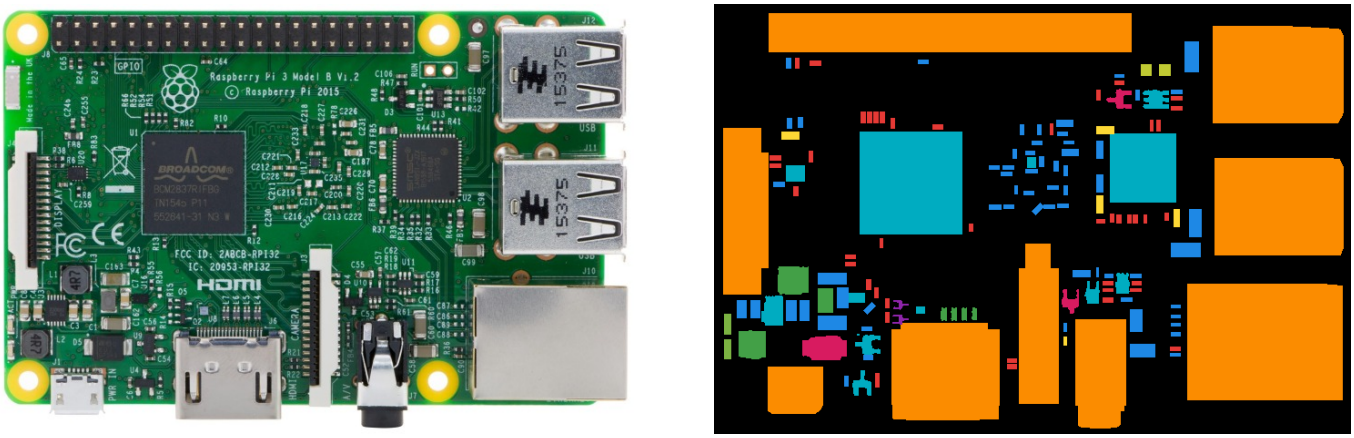
```
alias loginhpc='ssh -X -i /path/to/csisdl_XX csisdl_XX@hpc.bits-hyderabad.ac.in'
```

Use only your allocated HPC account and GPU time; store large data in permitted scratch storage. Kaggle Notebooks may be used instead, subject to its current GPU availability and quota. **Never run large downloads on a login node against local cluster policy.**

2. Background, Dataset, and the Meaning of Zero-Shot

Research source. C. Pramerdorfer and M. Kampel, *A Dataset for Computer-Vision-Based PCB Analysis*, MVA 2015, pp. 378–381; DOI: 10.1109/MVA.2015.7153209. Dataset DOI: 10.5281/zenodo.3886553.

Images	748 DSLR PCB photographs; 165 distinct physical boards; typically 3–5 orientations/images per PCB.
Resolution	4928 × 3280 pixels, Nikon D4 camera.
Provided labels	Binary *whole-PCB foreground masks*; bounding boxes for 9,313 IC appearances; text labels for a subset of ICs.
Not provided	**No official per-component 25-class semantic masks or instance masks are available.** Whole-board foreground is *not* a component annotation.



(a) Original RGB PCB example

(b) Color-coded instance masks on black

Figure 1: **Required visualization format** (instructor-provided Raspberry Pi example): original RGB beside class-colored component masks on a pure-black canvas. This is an *illustration*, not an original DSLR target image or independently verified ground truth. The class-color table below defines the authoritative palette.

Zero-shot cross-dataset transfer (this assignment): Train, select hyperparameters, and freeze your model using the *previously available labeled source-domain data* (Graph-W/Graph-F training data). Then apply the fixed system directly to the *new, unseen target domain*: the PCB DSLR dataset. **No DSLR component labels or DSLR teacher pseudo-labels may be used for student weight training, hyperparameter selection, or checkpoint selection in the graded zero-shot track.** Inference-time resizing/normalization is permitted. The Instructor may score against a *frozen YOLO26-L + SAM teacher reference* (described below). This is *unseen-domain zero-shot transfer*, not open-vocabulary zero-shot recognition or target-domain teacher distillation.

Important scope. Retain the original 25 class IDs, including the two legacy IC entries *U* and *IC*, and the two switch entries *SW* and *S*. Reuse the *same operational distinction/mapping/color-coding of 25 components* documented for Assignment–I. If ambiguous in DSLR images, predict *unknown/ignore* rather than inventing a new class. Background/void has ID 0 and is not a 26th target component class. A target class absent from a particular PCB image need not be predicted.

2.1 Official Zenodo download procedure

Download from `https://zenodo.org/records/3886553`. The record provides **eight separate ZIP archives**, `cvl_pcb_dslr_1.zip` through `cvl_pcb_dslr_8.zip`, totaling about **6.2 GB compressed**. Select each *Download* link, or use a terminal on an approved storage/compute node:

```
mkdir -p pcb_dslr_download && cd pcb_dslr_download
for i in $(seq 1 8); do
  curl -L --fail --retry 5 -C - \
    -o "cvl_pcb_dslr_${i}.zip" \
    "https://zenodo.org/records/3886553/files/cvl_pcb_dslr_${i}.zip?download=1"
done
for f in cvl_pcb_dslr_*.zip; do unzip -t "$f" >/dev/null || exit 1; done
for i in $(seq 1 8); do
  mkdir -p "part_${i}"
  unzip -q "cvl_pcb_dslr_${i}.zip" -d "part_${i}"
done
```

The official research API/example code is available at `https://github.com/cpra/mva15` (its Python API targets Python 2; students may use their own loader). Verify MD5 checksums listed on Zenodo, retain the source archive names, inspect extracted folders before reorganizing, document your final path layout, and cite the dataset. **Split by physical PCB identity**, not image filename: other viewpoints of the same board must not create train/validation or benchmark leakage. Do not use the DSLR IC boxes or PCB masks for training/model selection in the graded zero-shot setting; those annotations are reserved for permitted evaluation/proxy checks.

3. Pseudo-Ground-Truth Protocol and Model Budget

Goal. A single, frozen **YOLO26-L-seg + Segment Anything Model (SAM)** pipeline will produce *geometry-aware pseudo-instance masks* for evaluating compact student models. **Treat these as reference pseudo-labels, not human ground truth.** The same instructor-controlled teacher, thresholds, class mapping, mask resolution and evaluation images must be used for every group.

T1. 25-class detection. The YOLO26-L segmentation/detection head must first be *adapted using only the 25-class source data* from Assignment–I (or paired with the same frozen 25-class source-trained classifier). *Off-the-shelf COCO-pretrained YOLO26 cannot automatically name all 25 specialized PCB classes.* Retain each proposal's class ID, bounding box and confidence.

T2. SAM geometry refinement. Feed each retained component box (and, if justified, prompt points) to frozen SAM. Extract the corresponding high-resolution binary mask. **SAM supplies geometry, not a reliable PCB class ID**; inherit class identity only from the 25-class detector/classifier.

T3. Fixed annotation quality-control. Remove duplicate proposals with documented same-class mask-NMS, discard obvious PCB-background artifacts using the provided whole-board mask, resolve overlap consistently, preserve small resistor/capacitor instances and record uncertain regions as `ignore=255`. Never force a low-confidence guess into a 25-class label.

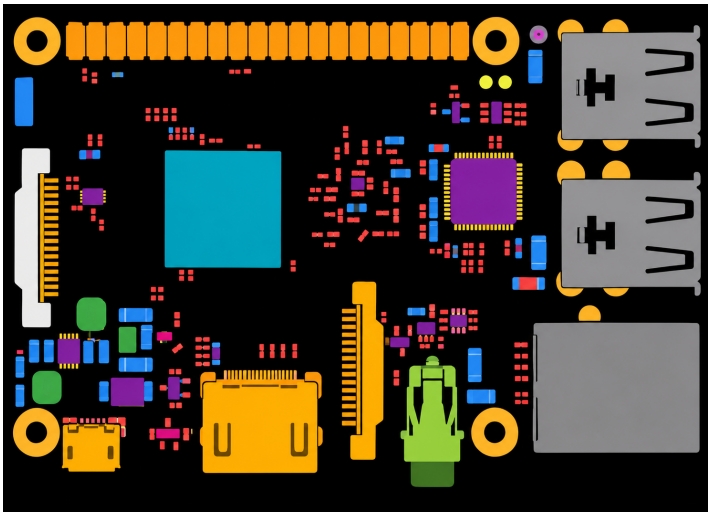
T4. Versioned scoring reference. Use `class_id`, `instance_id`, `bbox`, `score`, and encoded `mask` per detected component; unlabeled pixels use background 0 or ignored 255.

Permitted student use: Students may run YOLO26-L + SAM to *illustrate or diagnose* pseudo-masks on a separate, designated public visualization subset and may use this teacher on *source-domain* training data. However, a student's self-generated DSLR reference labels are **not accepted as that student's grading ground truth**. Teacher predictions on the Instructor's held-out DSLR evaluation set are reserved for scoring; no target-domain teacher-student fine-tuning or threshold tuning is allowed for the graded *zero-shot* submission. Target pseudo-label fine-tuning, if investigated, must be reported in a separate non-graded *pseudo-supervised adaptation* experiment.

Inference constraint. Your segmentation pipeline (*i.e.,* backbone, class and mask heads, any decoder, and any other learned auxiliary modules) must satisfy total parameters

$$P_i \;\le\; P_{\max} = P(\text{YOLO26m-seg}) \;\approx\; 23.6 \times 10^6 \text{ parameters.}$$

The **approximate 23.6M** value is the published *fused-inference* YOLO26-M-seg count (YOLO26-L-seg is approximately **28.0M**). The Instructor's pinned reference checkpoint and *same export/fusion convention* determine the exact ceiling. Count *all* weights needed at inference across the full model, not merely trainable weights, excluding only the detached offline teacher. Ensembles, SAM calls, or hidden classifiers during student inference **also count** and cannot be used to circumvent the cap.



Expected geometry-aware mask:

- Individual small passives remain separated.
- Need not be exact and accurate to the illustrated geometry, but as close as possible.
- Large chips and connectors have coherent interiors.
- All non-component pixels remain black.
- Identical-class instances share a class color but have distinct IDs.

Figure 2: Expected geometry-aware instance segmentation of PCB components on a black background.

4. Question 1 – Zero-Shot Semantic Segmentation [24 Marks]

Problem. Build a deep neural network that predicts a single *semantic* class label for each PCB image pixel: background/void 0, or one of the same 25 component classes. You must **reuse and improve your Assignment–I training pipeline**. The final trained model must be frozen before processing target DSLR images. A shared trained backbone, decoder, and experimental runs may be used for both questions.

(a) Method (required): Implement or adapt the *backbone → multi-scale pixel decoder → masked-attention Transformer decoder* pipeline of Mask2Former. *Re-engineer* at least **one substantive internal mechanism** (e.g., multi-scale fusion, masked-attention refinement, boundary-aware objective, number/selection of mask queries, compact backbone, class imbalance strategy). State what changed *in code* and why. Merely changing the classifier's class count is **not** sufficient.

(b) Zero-shot source training: Train on permitted labeled source data only; preserve the same 25-ID mapping. Predict all 25 channels even though some target-board classes may be absent. Document unknown/void handling and threshold decisions based only on source validation.

(c) Required comparisons: (i) best applicable Assignment–I model (ported where necessary); (ii) your improved new model. Use at least one controlled ablation and report total parameters, trainable parameters, inference time, memory, losses, and source-domain validation results.

(d) Explainability: Apply **Grad-CAM** to a differentiable class/mask score using a named spatial backbone/pixel-decoder feature layer. Save heatmaps on DSLR images; explain what regions drive component predictions and examine at least two errors/failures. Gradient maps from segmentation scores must not be confused with attention matrices.

(e) Outputs: Full-board semantic map, RGB overlay, pure-black canvas, prediction/confidence statistics, Grad-CAM, source/target qualitative comparison, complete inference-parameter count and teacher-reference *diagnostic* score on any designated public subset. Submit `best_semantic.ckpt` and executable standalone inference.

Q1 assessment item	**Marks**
Blinded semantic agreement with frozen, quality-audited DSLR teacher pseudo-labels	12
New Model, methodology and source-to-target justification	5
Controlled comparison, ablation, parameter/time efficiency and error analysis	4
Grad-CAM correctness and required semantic visualizations	3
TOTAL (Question 1)	**24**

5. Question 2 – Zero-Shot Instance Segmentation [30 Marks]

Problem. Extend the same system to predict a *set* of distinct components with a 25-class label, a confidence score, and a separate binary mask for *every instance*. **A semantic map, bounding boxes alone, or a generic whole-PCB mask do not meet this question's requirements.**

(a) Ablations: Compare the Assignment–I segmentation baseline and your final model under the same source-only training and frozen DSLR inference protocol. Show whether your model is affected or performs better on small-component segmentation, boundaries, cost, or false positives. You may reuse the three Q1 comparison runs when the shared model supports both tasks.

(b) Required imagery: For each evaluation sample, save (i) original board image, (ii) RGB overlay, (iii) colored non-overlapping *instance* masks on **black background**, with visible class colors, unique integer instance IDs, and (iv) a summary listing per-class counts/confidence. Do not hallucinate masks for classes that are not present.

(c) Grad-CAM: For at least five predicted instances over at least three DSLR boards, produce mask-conditioned Grad-CAM (or a rigorous segmentation-target equivalent); specify target mask/query, score function, hooked spatial layer, and at least two analysis observations.

(d) Evidence: On reference-scored images, report *pseudo-reference* instance F_1, AP^{ps}_{50} and $mAP^{ps}_{50:95}$; distinguish them explicitly from metrics on human-verified masks. Evaluate class-aware, one-to-one mask matches at IoU ≥ 0.5. Provide class-wise counts, precision, recall and threshold curves. Save `best_instance.ckpt`.

Q2 assessment item	**Marks**
Blinded instance-mask F1 and AP against fixed, audited teacher pseudo-labels	16
Your deep neural model re-engineering, matching, boundary/small-object strategies	6
Ablations, source-domain quantitative analysis, efficiency and error diagnosis	4
Grad-CAM, black-background mask renders, IDs and reproducible outputs	4
TOTAL (Question 2)	**30**

6. Evaluation and Relative Grading Using a Frozen Teacher

Reference benchmark. Since the original DSLR dataset supplies only PCB foreground masks and IC boxes, the Instructor's *frozen YOLO26-L + SAM pseudo masks* serve as a common **reference for relative comparison**. This

is more reproducible than permitting each group to score itself against its own generated masks. It does *not* measure true 25-class accuracy unless independently checked. The Instructor should manually inspect a small stratified set (e.g., 20–30 ROIs across different physical PCBs), fix or mark unreliable labels `ignore`, and use the same audited images for every group. Report the audit outcome separately as a reliability check. Do not credit unverified regions as perfect predictions.

Pixel-level semantic score. On teacher-supported, non-ignored components, calculate per-class precision P_k, recall R_k and $F_{1,k} = 2P_kR_k/(P_k + R_k)$. Define *macro F1* over supported classes, excluding absent/unlabeled classes, and class-macro $mIoU^{ps}$. The superscript *ps* means *pseudo-reference*, not human ground truth. Background pixels must contribute to false positives when your trained model hallucinates PCB components.

Instance-level score. At different confidence thresholds τ ($0.1, 0.2, ..., 1.0$ range) selected from *source-only validation*, perform maximum-IoU one-to-one matching of predicted masks to pseudo-reference instances *within the same class*; a TP requires mask-IoU ≥ 0.50. Unmatched predictions and pseudo-reference instances count as FP and FN. Define

$$F^{ps}_{1,\text{inst}} = \frac{2TP}{2TP + FP + FN} \quad (0 \leq F_1 \leq 1); \qquad F_1 = 1 \iff \text{perfect agreement under this matching rule.}$$

Compute AP^{ps}_{50} using scored/ranked predictions (do not confuse AP with a threshold-specific F1), and optionally $mAP^{ps}_{50:95}$. A score *greater than 1 is mathematically impossible.* Also report mask Dice/IoU for successfully matched instances to expose overly coarse shapes.

Task quality indices (the primary goal is to approach F1 = 1):

$$\boxed{\begin{aligned} C_{1,i} &= 0.75\, F^{ps}_{1,\text{sem},i} + 0.25\, mIoU^{ps}_i, \\ C_{2,i} &= 0.75\, F^{ps}_{1,\text{inst},i} + 0.25\, AP^{ps}_{50,i}. \end{aligned}}$$

Both scores lie in $[0, 1]$. To reward accurate *and compact* models, let P_i be the deployed inference parameters, $P_{\max}$ the pinned YOLO26-M-seg ceiling, and r_i the size of externally supplemented *labeled source-domain data* relative to the original source training size. Define

$$E_i = \text{clip}\left(\sqrt{\frac{P_{\max}}{P_i}},\ 1,\ 1.10\right), \quad A_i = \min\left(1, \frac{1.5}{r_i}\right), \quad U_{q,i} = C_{q,i}E_iA_i.$$

For $P_i > P_{\max}$, the final student model is **ineligible for benchmark marks**; its teacher does not count against the budget only if used strictly *offline*, never in student inference. Within the parameter ceiling, the benchmark marks for question q are

$$\boxed{M^{\text{benchmark}}_{q,i} = B_q\, C_{q,i} A_i \left(0.70 + 0.30\, \frac{U_{q,i}}{\max_{j \in \mathcal{V}} U_{q,j}}\right)}, \quad B_1 = 12,\ B_2 = 16,$$

where $\mathcal{V}$ is the set of parameter-compliant, valid group submissions and the relative term is defined as zero if all group utilities are zero. The remaining **Q1: 12**, **Q2: 14**, and **submission: 6** marks follow the question rubrics and checklist. The multiplicative factor ensures that a weak cohort cannot receive high performance marks merely by ranking first; the relative term rewards quality and compactness among groups. Smaller models receive up to a **10% utility advantage**, never compensation for clearly inaccurate segmentation.

Important grading interpretation: A high F^{ps}_1 means your final neural model reproduces *YoLoV26-generated* masks, including any mistakes. It must label scores as *pseudo-reference agreement*, inspect false positives and omissions on the audited ROIs, and reserve *human-GT accuracy* terminology for separately verified masks. Whole-PCB masks and published IC boxes remain optional auxiliary checks, not substitutes for class-aware component GT.

Note: No DSLR pseudo-mask training is permitted in the *zero-shot* track. Reuse Assignment–I weights, pre-processing, source-domain evaluation and existing inference scripts. The aim is comparable working hours despite the doubled formal marks.

7. Required Files, Naming, and Reproducibility Contract

Artifact	Required evidence	Exact name/path
Complete source code	Source-only training, Mask2Former architecture changes, matching/loss, evaluation, plotting, Grad-CAM. Cite external libraries and modifications.	`src/semantic/`; `src/instance/`
Best checkpoints	Both runnable frozen checkpoints; no manual test-time changes.	`best_semantic.ckpt`; `best_instance.ckpt`
Standalone inference	One independent Python file per task: model construction, forward, pre/post-processing, rendering and Grad-CAM inside; imports only installed third-party packages.	`inference_semantic.py`; `inference_instance.py`

Artifact	Required evidence	Exact name/path
Shell launchers	One-command reproduction using provided sample input, checkpoint, target output dir, and optional `-gradcam`; no import from training-time `src/` code.	`run_semantic.sh`; `run_instance.sh`
Target images	Original DSLR RGB; transparent overlay; *pure-black canvas with colored class masks*; instance outlines/IDs; Grad-CAM heatmaps; corresponding legend. Include the comparison to a released teacher mask when available.	`results/semantic/`; `results/instance/`
Plots	Source epoch losses; P/R/F1 versus thresholds [0,1] on source labels and any released pseudo-reference samples; mIoU/Dice, AP, Grad-CAM and ablation plots. Never report hidden teacher scores as student-computed.	`results/plots/`
Experiment JSON	Consistent keys: `run_id`, `task`, `seed`, `train_loss`, `val_loss`, `val_metrics`, `params_total`, `params_inference`, `params_trainable`, `runtime`, `data_source`, `threshold`, `checkpoint`, `teacher_version`; report `pseudo_metrics` only for public diagnostic data.	`semantic_stats.json`; `instance_stats.json`
Class mapping	Canonical 25 class IDs/colors, U-vs-IC and SW-vs-S handling, background 0, ignore 255, dataset-to-class mapping; versioned pseudo-reference schema.	`data_mapping.json`
Documentation	Non-expert guide to download/checksum, environment, source train/val, exact inference commands, parameter-count audit, teacher/zero-shot separation, Grad-CAM and evaluation caveats.	`README.md`; `config.yml`

Required naming for each board `<board_id>`:

`<board_id>_rgb.png` `<board_id>_overlay.png`
`<board_id>_black_masks.png` `<board_id>_gradcam.png`

Additionally save `<board_id>_instances.json` in Question 2. Preserve the common palette. Include the file `run_manifest.json` listing all tested board IDs, image dimensions, model hash, source-only threshold, command line, software versions, and execution timestamps.

8. Submission & Organization – 6 Marks (All-or-Nothing)

All 6 marks are awarded only when every mandatory item below is correct; otherwise the submission/organization component is 0/6. The question-specific marks are unaffected, subject to scientific validity and reproducibility.

- ☐ Correct group ownership and expected folders.
- ☐ Exactly one email from one group member; use exactly these identifiers:
  `[CSF425_2026_XX_Assignment-II]` (subject)
  `CSF425_2026_XX_Assignment-II.zip` (ZIP file).
- ☐ Both checkpoints and both shell/Python standalone inference pairs run from a fresh documented environment.
- ☐ Required black-background semantic and instance-mask figures, instance IDs, class legends, Grad-CAM outputs, and inference parameter-count report exist with exact naming.
- ☐ Valid `semantic_stats.json`, `instance_stats.json`, full ablation statistics, 25-class mapping, parameter compliance, and no falsely claimed human-GT DSLR metrics.
- ☐ README fully reproduces download, source training, inference, plots and model scoring; complete citations; no leaked credentials, unlicensed redistribution or hidden target-label tuning.

9. References and Required Reading

1. Pramerdorfer, C., and Kampel, M. (2015). *A Dataset for Computer-Vision-Based PCB Analysis*. MVA, 378–381. https://doi.org/10.1109/MVA.2015.7153209. *Attached paper is the primary dataset reference.*
2. PCB DSLR Dataset (Zenodo v1, 2020). https://zenodo.org/records/3886553. Eight archives and official archive MD5 values; non-commercial research use.
3. Dataset code and access API: https://github.com/cpra/mva15.
4. Cheng, B. et al. (2022). *Masked-Attention Mask Transformer for Universal Image Segmentation*. CVPR, 1290–1299. https://doi.org/10.1109/CVPR52688.2022.00135.
5. Mask2Former reference implementation: https://github.com/facebookresearch/Mask2Former. Code reuse requires attribution and a documented architectural ablation.
6. Ultralytics (YOLO26) official instance-segmentation reference and published YOLO26-M / YOLO26-L parameter counts: https://docs.ultralytics.com/tasks/segment/. Baseline weights are COCO-pretrained; 25-class adaptation is required.
7. Kirillov, A. et al. (2023), *Segment Anything*. ICCV. https://github.com/facebookresearch/segment-anything. SAM refines masks from prompts; it does not automatically assign the specialized PCB class taxonomy.